

Adam **Dennett**

Professor of Urban Analytics

Bartlett Centre for Advanced Spatial Analysis, University College London

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Profile

I am Professor of Urban Analytics at the Bartlett Centre for Advanced Spatial Analysis (CASA), University College London. My research uses data and quantitative methods — frequently in R — to understand cities, migration, housing and the systems people create. I publish across migration modelling (particularly spatial interaction models), housing and energy, gentrification, synthetic data and the geographies of beer and brewing, and have a passion for applied work in urban policy contexts.

Professional History

Professor of Urban Analytics
Bartlett Centre for Advanced Spatial Analysis, UCL — *2021–Present*

Head of Department and Director
Bartlett Centre for Advanced Spatial Analysis, UCL — *2018–2023*

Associate Professor, Urban Analytics
Bartlett Centre for Advanced Spatial Analysis, UCL — *2017–2021*

Deputy Director
Bartlett Centre for Advanced Spatial Analysis, UCL — *2016–2018*

Lecturer in Urban Analytics
Bartlett Centre for Advanced Spatial Analysis, UCL — *2013–2017*

Post-doctoral Research Associate
Bartlett Centre for Advanced Spatial Analysis, UCL — *2010–2013*

Research Assistant and part-time PhD student
School of Geography, University of Leeds — *2006–2010*

Teacher of Geography and Head of General Studies
Hedingham School, Essex — *2002–2005*

Education

PhD in Quantitative Human Geography
School of Geography, University of Leeds — *2007–2011*

- Studied part-time

MA in Human Geography with Geographic Information Systems
School of Geography, University of Leeds — *2005–2006*

- Pass with Distinction

Qualified Teacher Status (QTS)
National College of Teaching and Leadership, DfE — *2003–2003*

PGCE in Secondary Geography
Faculty of Education (Homerton College), University of Cambridge — *2001–2002*

BSc (Hons) Geography
Lancaster University — *1998–2001*

Appointments and Affiliations

Member, ESRC Peer Review College

ESRC *2024–Present*

External Examiner, MSc Geographical Information Science

University of Manchester *2023–2027*

- Department of Geography, School of Environment, Education and Development, Faculty of Humanities
- Appointed 1 October 2023 to 31 December 2027

Editor-in-Chief, Applied Spatial Analysis and Policy (Impact Factor 1.9)

Springer *2017–2023*

Fellow of the Academy of Social Sciences (FAcSS) — Elected

Academy of Social Sciences *2021–Present*

Fellow of the Royal Geographical Society (FRGS)

Royal Geographical Society *2010–Present*

Prizes, Awards and Honours

Best Paper, 33rd Annual GIS Research UK (GISRUK) Conference

University of Bristol *2025*

Elected Fellow of the Academy of Social Sciences (FAcSS)

Academy of Social Sciences *2021*

Cabinet Office Geospatial Commission, Geography in Government Award — Excellence in Local Government

Awarded with City of Westminster Council for pioneering Urban Lab work *2020*

Best Paper, 27th Annual GIS Research UK (GISRUK) Conference (Colouring London Project)

University of Newcastle *2019*

UCLU Student Choice Teaching Award for Outstanding Teaching

University College London *2016*

Grebenik Prize — best work on any aspect of human population problems

University of Leeds *2006*

Grants

External grants total awarded as PI or Co-I since 2012: **£17,491,621.90**.

Census Innovation at CeLSIUS (ES/Z502741/1)

Co-I; active *2024*

- £384,576

TRACK-UK: Synthesized Census and Small Area Statistics for Transport and Energy (ES/Z50290X/1)

Co-I; active *2024*

- £327,680

AI for Collective Intelligence (AI4CI) — UKRI AI Research Hub for Real Data (EP/Y028392/1)

Co-I; Local PI for UCL branch; active *2024*

- £10,274,279 (UCL portion £980,391.77)

Pathways to socially equitable net-zero cities and regions — ESRC CASE PhD Studentship

PI/Supervisor (Transport East partner); active *2023*

- £93,633.90

Equity Assessment of Demand-side Flexibility Technologies in Grid Integrated Transport Electrification (PhD)

PI/Supervisor (EPSRC CREDS); active *2023*

- £96,044

Generative scenarios for getting Transport to an equitable Net Zero - flipping modelling on its head using Machine Learning (2868504)

active *2023*

- EPSRC

NARRATE

PI; complete *2022*

- £142,739.24 (Ordnance Survey)

dunnhumby PhD Studentship

PI/Supervisor (industry-funded); complete *2020*

- £60,000 (dunnhumby)

CeLSIUS (ES/V003488/1)

Co-I; active *2020*

- £2,160,348.40 (ESRC)

YOUR.VU.CITY (UKRI 105325)

Local PI (Co-I on whole project); complete *2019*

- £620,697 (UCL £159,996; Innovate UK)

CeLSIUS Re-commissioning (ES/R00823X/1)

Co-I; complete *2018*

- £1,020,753 (ESRC)

Colouring London Impact Acceleration Grant

PI; complete *2018*

- £55,200 (EPSRC)

CeLSIUS Extension Grant

Co-I; complete *2016*

- £247,182.40 (ESRC)

Colouring London

PI (with Co-PI Polly Hudson); complete *2016*

- £58,400.74 (Historic England)

Understanding London's Resident Metabolism (1827316)

complete *2016*

- EPSRC

RGS Innovative Teaching Grant

PI (Co-I Paul Turner, Bedales School); complete *2014*

- £1,000 (RGS)

SYLLS — Synthetic data estimation for the UK Longitudinal Studies

PI; complete *2013*

- £272,652.73 (ESRC)

CeLSIUS (ES/K000365/1)

Co-I; complete *2012*

- £1,676,436 (ESRC)

UK DATA SERVICE-CENSUS SUPPORT SERVICE (UKDS-CSS) (ES/J02337X/1)

complete *2012*

- £2,011,044 (ESRC)

Invited Talks (recent)

Department of Real Estate and Planning, Henley Business School, University of Reading
Can we model our way out of this? — Putting the AI in Applied Urban Modelling. AI, collective intelligence, and the Brighton & Hove school system — *18 Nov 2026 (upcoming)*

Geospatial Data Science Seminar, University of Glasgow
Can we model our way out of this? — Putting the AI in Applied Urban Modelling. AI, collective intelligence, and the Brighton & Hove school system — *13 Nov 2026 (upcoming)*

Brighton AI (meetup)
Can we model our way out of this? — Putting the AI in Applied Urban Modelling. AI, collective intelligence, and the Brighton & Hove school system — *4 Nov 2026 (upcoming)*

BHASVIC (Brighton & Hove Sixth Form College)
How to Pull the Right Lever — *July 2026*

UCL Development Planning Unit — talk to Latin American city leaders
Smart Cities and Collective Intelligence — *Sept 2025*

Royal Geographical Society Annual Conference
GIS vs The City Council: Data Analytics, Collective Intelligence and Policy Dust-ups (GISRUK 2025 Best Paper) — *Aug 2025*

Brighton Data Forum (public lecture)
The Case of the Brighton Secondary School Admissions Consultation — *March 2025*

Institute for Spatial Data Science, University of Leeds (departmental seminar)
Data Analytics and the Brighton Secondary School Admissions debate — *Feb 2025*

Parent Support Group, Brighton (online public lecture)
Data and evidence missing from the B&H Secondary School Admissions Consultation — *Jan 2025*

Alan Turing Institute / University of Glasgow
Early career researcher panel, Urban Analytics meeting — *Dec 2024*

Aalto University, Finland
Housing and Net-Zero — *Sept 2024*

Geographic Data Science Lab, University of Liverpool (departmental seminar)
Evidence against the effects of schools on local house prices in Brighton and Hove — *Feb 2024*

Carto Spatial Data Science Bootcamp, London (invited back)
Foundations of Geospatial — *Nov 2023*

Carto annual conference, Royal Geographical Society, London
Careers panel — *May 2023*

Carto Spatial Data Science Bootcamp, London
Foundations of Geospatial — *Feb 2023*

Teaching

Postgraduate

Course Director, MSc Smart Cities and Urban Analytics
Bartlett Centre for Advanced Spatial Analysis, UCL — *2013 – 2018*

CASA0007 Quantitative Methods
Teaching contributor — *2025/26 – Present*

CASA0001 Urban Systems Theory
Co-convenor (with Dr Esra Suel) — *2024/25 – Present*

CASA0008 Smart Cities: Context, Policy and Government

Co-convenor (with Dr Ed Manley) *2015/16 – 2016/17*

- Rebuilt from a guest-lecture series into a module we led and taught, with a student-built class wiki and flipped-learning group work
- Moved from among the weakest to among the best student feedback in the department

CASA0001 Urban Systems Theory

Co-convenor (with Dr Elsa Arcaute) *2014/15 – 2017/18*

- Core foundational module; wrote and taught the City Demography and City Inequalities sessions
- Grew from 28 to 55 students by 2016/17

BENVGSC6 Dissertation

Co-convenor, dissertation preparation classes (with Dr Martin Austwick, Dr Elsa Arcaute) *2014/15 – 2016/17*

- Wrote the CASA dissertation handbook guiding students through the dissertation process

GEOGG201 Interdisciplinary Approaches to the Study of Migration

Guest lecturer, UCL Geography (c.30 students) *2014/15*

- Lecture on quantitative migration modelling, drawing on my own research

CASA0005 Geographic Information Systems and Science

Module convenor; core module for both CASA MSc programmes *2013/14 – 2022/23*

- Grew from 25 to as many as 150 students, with taught content and assessment extensively reworked
- Replaced the unseen written exam with an oral examination, and added research-led practicals (UCL Connected Curriculum)
- Coursework by a 2014/15 student was developed and published in Transactions in GIS

Undergraduate

BASC2042 Understanding Cities and Spatial Cultures

Lecture on city populations and inequalities, UCL Arts and Sciences (c.30 students) *2014/15 – 2015/16*

GEOG1004 Human Ecology

Five lectures on a core undergraduate geography course, UCL Geography (c.100 students) *2011/12*

PhD Supervision

Current research students

1st supervisor

Yujing (Olivia) Xing — Thesis title to be confirmed. *2026*

1st supervisor

Maria Wood — Equity assessment of demand-side flexibility technologies in transportation decarbonisation (with Esra Suel, Gerry Casey). Upgrade passed. *2023*

1st supervisor

Claude Lynch — Pathways to Socially Equitable Net Zero Cities and Regions (with Esra Suel, Gerry Casey). Upgrade passed. *2023*

1st supervisor

Tom Murat — Generative scenarios for getting Transport to an equitable Net Zero (with Esra Suel, Gerry Casey). *2023*

1st supervisor

Denise Mizzi (PT) — Small Area Estimation of Wheezing prevalence in Maltese School Children (with Jens Kandt, Nicola Shelton). Upgrade passed. *2018*

2nd supervisor

Roei Yosifof — Application of novel AI methods in the Digital Urban Planning domain. AI4CI Linked PhD. *2025*

2nd supervisor

AI4CI Linked PhD (Glasgow half of Cities Theme) — candidate appointed. *2025*

2nd supervisor

Chiara Gericke — Role of the Neighbourhood Food Environment in Susceptibility to Obesity (1st sup. Claire Llewellyn, Institute of Epidemiology and Health Care). *2023*

Completed

1st supervisor

Philip Wilkinson — Applications of novel retail modelling methodologies in an evolving grocery sector (2nd sup. Neave O'Clery). Minor corrections. *2020 – 2023*

1st supervisor

Bonnie Buyuklieva — Understanding London's Residential Metabolism (2nd sup. Hannah Fry). Minor corrections. *2016 – 2022*

1st supervisor

Bin Chi — Spatial and temporal patterns of residential house prices and affordability in the UK (2nd sup. Thomas Oléron Evans). Passed without corrections. *2016 – 2021*

1st supervisor

Matthew Ng — The future of Brunei: changes in Brunei-Muara's urban and population landscapes (2nd sup. Elsa Arcaute). Minor corrections. *2016 – 2020*

2nd supervisor

Yufeng Wu — Dynamics of return migration intention and behaviour in China (1st sup. Rory Coulter). *2019 – 2024*

2nd supervisor

Lixun Liu — Impacts of rail transit investment on development in Chongqing (1st sup. Robin Hickman). Minor corrections. *2015 – 2016*

MSc / MRes

Supervised 45+ dissertations between 2017–2025. Joshua Yee (2019) won the RGS-IBG Population Geography Research Group postgraduate dissertation prize.

Research Staff Line Management

Dr Beatrice Taylor; Adham Enaya; Dr Yahya Gamal (Glasgow — oversight)

AI for Collective Intelligence National Research Hub *2024-Present*

Dr Bonnie Buyuklieva (now Lecturer in Digital Humanities, UCL)

NARRATE — Ordnance Survey Housing Net Zero R&D Project *2022-2023*

Dr Martin de Jode (now Lecturer in CASA); Maciej Ziarkowski; Dominic Humphrey (now UCL Energy Institute)

Colouring London (CASA) *2018-2020*

Dr Belinda Wu

SYLLS *2013-2014*

Academic Staff Line Management

Currently line-manage 2 academic staff. As Head of Department over a 5-year period directly line-managed (seeing several through promotions):

- 3 Professors (one successfully re-banded during period)
- 6 Associate Professors (2 promoted to Professor during period)
- 4 Lecturers (2 promoted to Associate Professor during period)

Selected Publications (since 2020)

A full list is available via Google Scholar and ORCID 0000-0001-8036-0185.

Gericke, C., Llewellyn, C., Burgoine, T., Dennett, A.
Clustering of takeaway outlets around London schools: A geospatial analysis. *Health and Place*. doi:10.1016/j.healthplace.2026.103753 — *2026*

Lynch, C.E.H., Casey, G., Suel, E., Dennett, A.
Is transport modelling 'a bit of a racket'? On the role of transport planning support systems in England. *Transportation Research Interdisciplinary Perspectives*. doi:10.1016/j.trip.2026.101876 — *2026*

Wood, M., Dennett, A., Casey, G., Suel, E.
Car dependence in England and Wales: spatial inequalities and implications for a just transition. *Frontiers in Sustainable Cities*. doi:10.3389/frsc.2025.1719495 — *2026*

Thies, S., Kennedy, C., Dennett, A., Stockton, J., Hurst, J., Taylor, J., ...
Commuting on the London Underground and Long-term Health Outcomes: A Cohort Study Using Externally Linked Census Data. *International Journal of Population Data Science*. doi:10.23889/ijpds.v11i5.3705 — *2026*

Horsfall, L., Kennedy, C., Shelton, N., Taylor, J., Hurst, J.R., Loxham, M., ..., Dennett, A.
High-resolution mapping of residential wood burning heat sources using Energy Performance Certificates: A case study of England and Wales. *Environment International*. doi:10.1016/j.envint.2025.109537 — *2025*

Buyuklieva, B., Bevilacqua, I., Dennett, A., Reades, J., Hubbard, P.
Life for rent: Evolving residential infrastructure in London and the rise of Build-to-Rent. *Urban Studies*. doi:10.1177/00420980241277684 — *2025*

Riedl, C., Cremer, D.D., Lucarelli, G., Antoine-Souklaye, E., Bullock, S., Ajmeri, N., ..., Dennett, A.
The potential and challenges of AI for collective intelligence. *Collective Intelligence*. doi:10.1177/26339137241308821 — *2025*

Tan, J., Dennett, A., Goodspeed, R., Suel, E., Chen, H., Barros, J., ...
Do Good Schools Influence Property Prices? Disentangling School and Neighbourhood Effects with Empirical Evidence from Brighton. *Urban Book Series*. doi:10.1007/978-3-031-98300-9_17 — *2025*

Bullock, S., Ajmeri, N., Batty, M., Black, M., Cartlidge, J., Challen, R., ..., Dennett, A.
Artificial Intelligence for Collective Intelligence: A National-Scale Research Strategy. doi:10.48550/arxiv.2411.06211 — *2024*

Gericke, C., Llewellyn, C., Burgoine, T., Dennett, A.
From textbooks to takeaways: a geospatial analysis of clustering of takeaway outlets near London schools. *The Lancet*. doi:10.1016/s0140-6736(24)02017-8 — *2024*

Suel, E., Lynch, C., Wood, M., Murat, T., Casey, G., Dennett, A.
Measuring transport-associated urban inequalities: Where are we and where do we go from here? *Transport Reviews*. doi:10.1080/01441647.2024.2389800 — *2024*

Rowe, F., Lovelace, R., Dennett, A., Wolf, L.J., Harris, R., Heppenstall, A.
Spatial interaction modelling: A manifesto. doi:10.4337/9781802203233 — *2024*

Buyuklieva, B., Oléron-Evans, T., Bailey, N., Dennett, A.
Variations in domestic energy efficiency by property, neighbourhood and local authority type: where are the largest challenges for the net-zero transition of the UK's residential stock? *Frontiers in Sustainability*. doi:10.3389/frsus.2024.1329034 — *2024*

Dennett, A., Gray, S., Wyszomierski, J., Hoalst-Pullen, N., Patterson, M.
From the Beer Orders to Last Orders - Legislation, Taxation and the Modern Beer Landscape of London. doi:10.1007/978-3-031-39008-1_3 — *2023*

Yan, X., Dennett, A.
Re-defining Transport for London's strategic neighbourhoods from spatial and social perspectives. *Applied Geography*. doi:10.1016/j.apgeog.2023.103116 — *2023*

Wu, Y., Coulter, R., Dennett, A.
Understanding the relationships between the family structures and destinations of married migrants with children in China. *Applied Geography*. doi:10.1016/j.apgeog.2023.103102 — *2023*

Liu, L., Dennett, A., Hickman, R.
Exploring the spatial differences in travel mode choice of rail transit in Chongqing. *Transportation Planning and Technology*. doi:10.1080/03081060.2023.2228301 — *2023*

Dennett, A., Buyuklieva, B.
Transitioning the residential environment of the UK to net-zero: An in-depth analysis of the energy performance of the building stock and its local social and spatial contexts. — *2023*

Buyuklieva, B., Dennett, A., Bailey, N., Morley, J.
Energy Efficient Homes: The Social and Spatial Patterns of Residential Energy Efficiency in England. *GISRUK proceedings*. doi:10.5281/zenodo.7855112 — *2023*

Liu, L., Dennett, A., Hickman, R.
Social equity impacts of rail transit investments using community perceptions: evidence from Chongqing. *Transportation Planning and Technology*. doi:10.1080/03081060.2022.2152454 — *2023*

Yee, J., Dennett, A.
Stratifying and predicting patterns of neighbourhood change and gentrification: An urban analytics approach. *Transactions of the Institute of British Geographers*. doi:10.1111/tran.12522 — *2022*

MacLachlan, A., Dennett, A.
An Applied Geographic Information Systems and Science Course in R. *Journal of Open Source Education*. doi:10.21105/jose.00141 — *2022*

Spooner, F., Abrams, J.F., Morrissey, K., Shaddick, G., Batty, M., Milton, R., ..., Dennett, A.
A dynamic microsimulation model for epidemics. *Social Science & Medicine*. doi:10.1016/j.socscimed.2021.114461 — *2021*

Clifford, B., Dennett, A., Chi, B.
Mapping Class E: Understanding the expansion of permitted development. — *2021*

Chi, B., Dennett, A., Oléron-Evans, T., Morphet, R.
A new attribute-linked residential property price dataset for England and Wales, 2011–2019. *UCL Open Environment*. doi:10.14324/111.444/ucloe.000019 — *2021*

Chi, B., Dennett, A., Oléron-Evans, T., Morphet, R.
Delineating the Spatio-Temporal Pattern of House Price Variation by Local Authority in England: 2009 to 2016. *Geographical Analysis*. doi:10.1111/gean.12287 — *2021*

Chi, B., Dennett, A., Evans, T., Morphet, R.
Shedding new light on residential property price variation in England: A multi-scale exploration. *Environment and planning. B: Urban analytics and city science*. doi:10.1177/2399808320951212 — *2020*